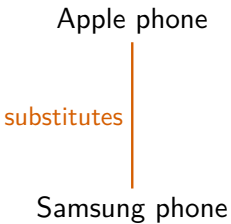
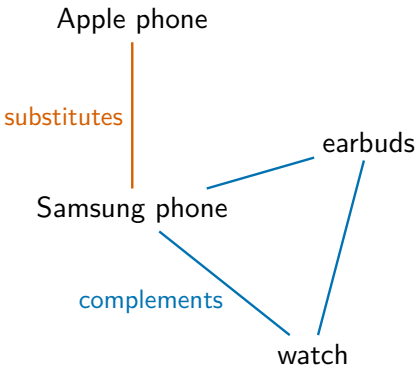
Robust Market Interventions

Andrea Galeotti (LBS)
Benjamin Golub (Northwestern)
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Eduard Talamàs (IESE)
Omer Tamuz (Caltech)

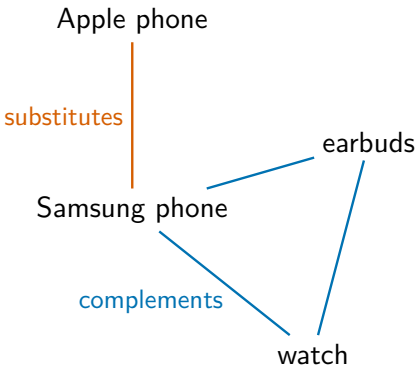
markets with many interacting goods



markets with many interacting goods

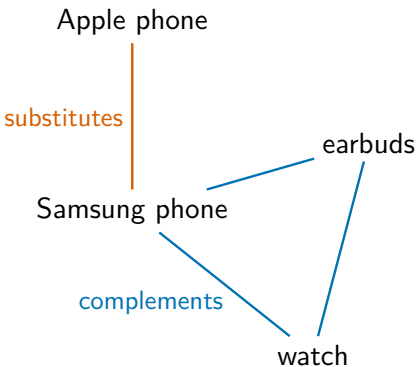


markets with many interacting goods



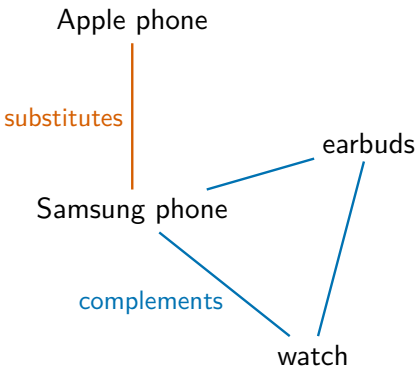
- spillovers across markets: lower phone prices increase demand for earbuds

markets with many interacting goods



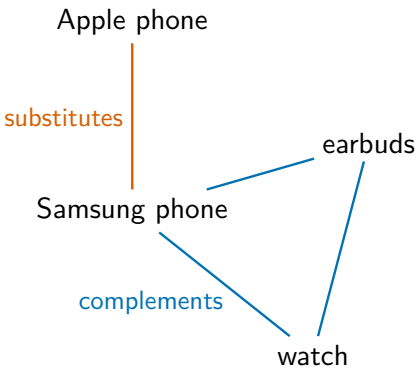
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- an authority (e.g., Amazon or a government) can tax or subsidize each good to raise total surplus

markets with many interacting goods



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markets with many interacting goods



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- an authority (e.g., Amazon or a government) can tax or subsidize each good to raise total surplus
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when do noisy estimates permit surplus gains with high probability in every market?

the answer

robust surplus gains depend on **one market property**

significant structure:
roughly, large groups of complementary goods

the answer

robust surplus gains depend on **one market property**

significant structure:
roughly, large groups of complementary goods

with this property

- a rule using only noisy estimates reliably approaches the complete-information optimum
- each $1 spent yields about $2 of market surplus
- consumers essentially unaffected

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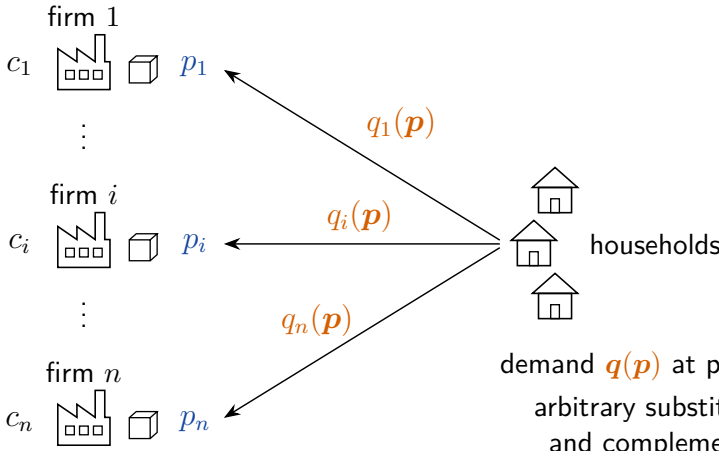
- in some markets, no rule guarantees even a net gain

significant structure ensures both *statistical* and *economic* properties,
and their interaction is interesting

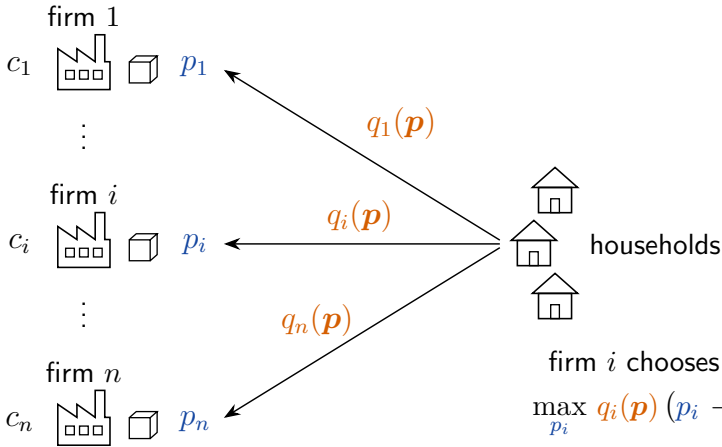
part I:

the oligopoly game

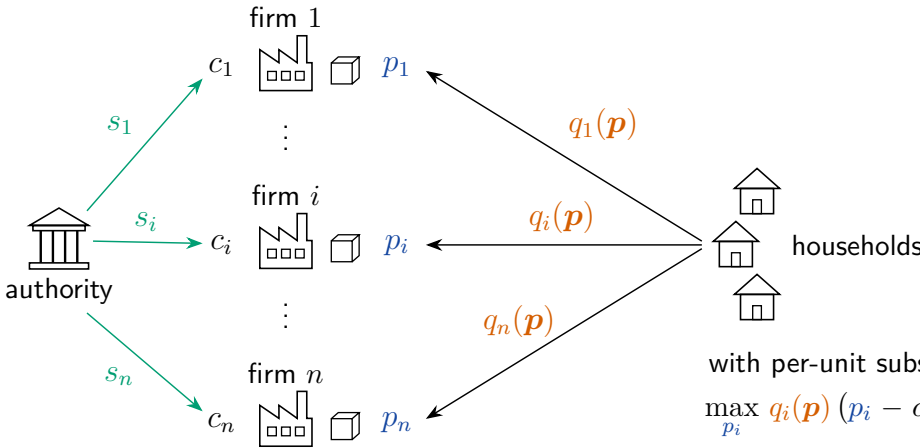
the game


firm 1
c_1
p_1
$q_1(\boldsymbol{p})$
firm i
c_i
p_i
$q_i(\boldsymbol{p})$
households
$q_n(\boldsymbol{p})$
firm n
c_n
p_n
demand $\boldsymbol{q}(\boldsymbol{p})$ at prices $\boldsymbol{p}$:
arbitrary substitutes
and complements

the game


firm 1
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p_i
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households
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firm n
c_n
p_n
firm i chooses p_i:
$\max_{p_i} q_i(\boldsymbol{p})\,(p_i - c_i)$
Nash equilibrium $(\boldsymbol{p}^0, \boldsymbol{q}^0)$:
status quo

the game



with per-unit subsidy s_i:

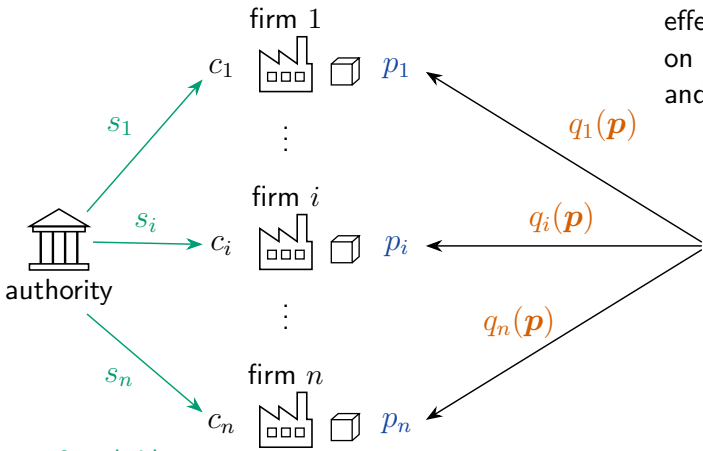
$$\max_{p_i} \; q_i(\boldsymbol{p}) \, (p_i - c_i + s_i)$$

effective marginal cost: $c_i - s_i$

$s_i > 0$: subsidy

$s_i < 0$: tax

the game



effect of a small subsidy $\boldsymbol{s}$
on equilibrium prices
and quantities

households

with per-unit subsidy s_i:

$$\max_{p_i} \; q_i(\boldsymbol{p}) \, (p_i - c_i + s_i)$$

effective marginal cost: $c_i - s_i$

$s_i > 0$: subsidy
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impact on equilibrium

demand responses at the status quo:

$$D_{ij} = \frac{\partial q_i}{\partial p_j}(\boldsymbol{p}^0)$$

demand matrix $\boldsymbol{D}$:
symmetric, negative semidefinite

$D_{ij} > 0$: **substitutes**
$D_{ij} < 0$: **complements**

units: $D_{ii} = -1$

the math →

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a small subsidy vector $\boldsymbol{s}$ changes equilibrium prices and quantities:

$$\Delta \boldsymbol{p} = -(\boldsymbol{I} - \boldsymbol{D})^{-1}\boldsymbol{s}, \quad \Delta \boldsymbol{q} = \boldsymbol{D}\,\Delta \boldsymbol{p}$$

$\Delta \boldsymbol{p}$, $\Delta \boldsymbol{q}$: first-order changes in prices and quantities

the math →

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assumption: near the status quo, each good's demand is linear in its own price

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the inverse incorporates cross-market spillovers: each of its entries depends on every entry of $\boldsymbol{D}$

assumption: near the status quo, each good's demand is linear in its own price

the math →

welfare

- **consumer surplus** CS: households' value of purchases minus expenditure
  $\mathrm{CS} = \sum_h \left[U^h(\boldsymbol{q}^h) - \boldsymbol{p} \cdot \boldsymbol{q}^h \right]$
- **producer surplus** PS: firms' profits, including subsidies
  $\mathrm{PS} = \sum_i \left(p_i - c_i + s_i \right) q_i$
- **spending** $S = \boldsymbol{s} \cdot \boldsymbol{q}$: subsidies times quantities sold
- **total surplus** $W = \mathrm{CS} + \mathrm{PS} - S$

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recall: impact on equilibrium

$$\Delta \boldsymbol{p} = -(\boldsymbol{I} - \boldsymbol{D})^{-1} \boldsymbol{s} \qquad \text{prices}$$

$$\Delta \boldsymbol{q} = \boldsymbol{D}\, \Delta \boldsymbol{p} \qquad \text{quantities}$$

market power implies underproduction; at status-quo quantities $\boldsymbol{q}^0$,

$$\Delta W = \boldsymbol{q}^0 \cdot \Delta \boldsymbol{q} = \sum_i \text{markup}_i \, \Delta q_i$$

ΔW: first-order total-surplus gain; in our units, $\text{markup}_i = q_i^0$

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output gains are weighted by markups; price changes are transfers

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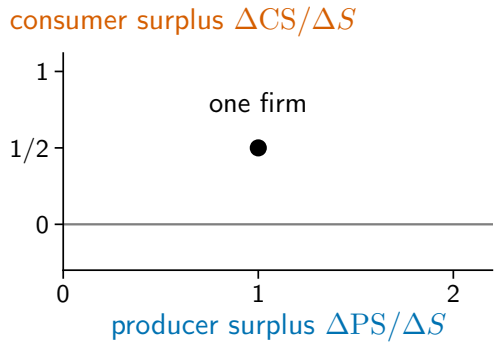
output gains are weighted by markups; price changes are transfers

goal: expand high-markup output

harder to achieve it: each subsidy moves every price and quantity

complete information

benchmark: one firm, linear demand



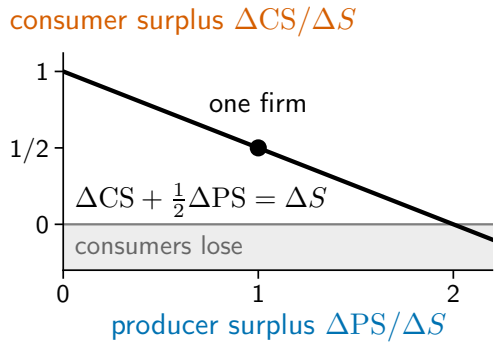
the math →

complete information

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Proposition 1: for every subsidy vector s, in every market,

$$\Delta\mathrm{CS} + \tfrac{1}{2}\,\Delta\mathrm{PS} = \Delta S$$



the math →

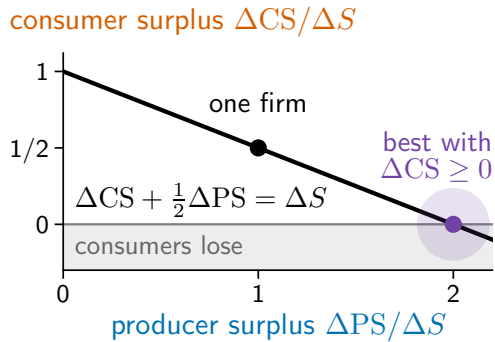
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complete information

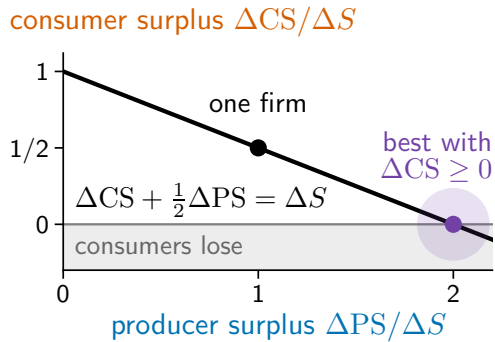
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with interacting goods, attaining **the corner** requires the full demand matrix $\boldsymbol{D}$



the math →

the problem

choose subsidies $\boldsymbol{s}$, with a fixed budget,
to make the *true* gain large:

$$\Delta W(\boldsymbol{s}) \;=\; \underbrace{(\boldsymbol{q}^0)^{\mathsf{T}}}_{\text{markups}} \underbrace{\left[-\boldsymbol{D}\,(\boldsymbol{I}-\boldsymbol{D})^{-1}\right]}_{f(\boldsymbol{D})} \boldsymbol{s}$$

$f(\boldsymbol{D})$: how quantities respond to $\boldsymbol{s}$

recall: where ΔW comes from

$\Delta \boldsymbol{p} = -(\boldsymbol{I}-\boldsymbol{D})^{-1}\boldsymbol{s}$	prices	$\Delta W = \boldsymbol{q}^0 \cdot \Delta \boldsymbol{q}$	total surplus
$\Delta \boldsymbol{q} = \boldsymbol{D}\,\Delta \boldsymbol{p}$	quantities	$\text{markup}_i = q_i^0$	in our units

the authority's problem

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the tempting shortcut

plug in $\widehat{\boldsymbol{D}}$ for $\boldsymbol{D}$ and act as if the estimate were right:

$$\widehat{\Delta W}(\boldsymbol{s}) \;=\; \underbrace{(\widehat{\boldsymbol{q}}^0)^{\mathsf{T}}}_{\text{estimated}} \underbrace{\left[-\widehat{\boldsymbol{D}}\,(\boldsymbol{I}-\widehat{\boldsymbol{D}})^{-1}\right]}_{f(\widehat{\boldsymbol{D}})} \boldsymbol{s}$$

the same function, evaluated at $\widehat{\boldsymbol{D}}$

recall: where ΔW comes from

$\Delta \boldsymbol{p} = -(\boldsymbol{I}-\boldsymbol{D})^{-1}\boldsymbol{s}$	prices	$\Delta W = \boldsymbol{q}^0 \cdot \Delta \boldsymbol{q}$	total surplus
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the authority's problem

the problem

choose subsidies s, with a fixed budget, to make the *true* gain large:

$$\Delta W(s) \;=\; \underbrace{(q^0)^{\mathsf{T}}}_{\text{markups}} \underbrace{\left[-D\,(I-D)^{-1}\right]}_{f(D)}\, s$$

$f(D)$: how quantities respond to s

the catch: D is unknown; the authority sees only a noisy estimate $\widehat{D} = D + E$, with errors as large as own-price effects

the tempting shortcut

plug in $\widehat{D}$ for D and act as if the estimate were right:

$$\widehat{\Delta W}(s) \;=\; \underbrace{(\hat{q}^0)^{\mathsf{T}}}_{\text{estimated}} \underbrace{\left[-\widehat{D}\,(I-\widehat{D})^{-1}\right]}_{f(\widehat{D})}\, s$$

the same function, evaluated at $\widehat{D}$

but s is scored by the *true* ΔW, and $f(\widehat{D})$ can be far from $f(D)$: the inverse can even flip the sign

recall: where ΔW comes from

$\Delta p = -(I-D)^{-1}s$	prices	$\Delta W = q^0 \cdot \Delta q$	total surplus
$\Delta q = D\,\Delta p$	quantities	$\text{markup}_i = q_i^0$	in our units

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the same function, evaluated at $\widehat{\boldsymbol{D}}$

but $\boldsymbol{s}$ is scored by the *true* ΔW, and $f(\widehat{\boldsymbol{D}})$ can be far from $f(\boldsymbol{D})$: the inverse can even flip the sign

a simple **control problem**: choose $\boldsymbol{s}$ for the true $\boldsymbol{D}$,
but $\boldsymbol{D}$ is seen only through $\widehat{\boldsymbol{D}}$: we need a **robust** control

related literature: markets and networks

Market power across markets and pass-through

Azar and Vives (2021); Pellegrino (2021);
Ederer and Pellegrino (2021); Weyl and Fabinger (2013)

Cross-market inefficiencies and cost pass-through. We study small taxes and subsidies that increase total surplus when cross-price effects are estimated imprecisely.

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Interventions in network games

Ballester, Calvó-Armengol and Zenou (2006); Galeotti, Golub and Goyal (2020)

Existing targeting results assume known spillovers. We derive spectral pass-through in oligopoly and target demand directions that remain identifiable under noise.

related literature: inference and decisions

High-dimensional statistics

Davis and Kahan (1970); Chen, Chi, Fan and Ma (2021)

Spectral methods recover common structure in a noisy matrix. We recover demand directions with high welfare returns to subsidies, without estimating every interaction.

related literature: inference and decisions

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Statistical decision theory

Wald (1950); Manski (2004); Kitagawa and Tetenov (2018); Athey and Wager (2021)

We require a high-probability welfare guarantee in every market state, with probability over the noisy signal and no prior over states.

part II: the main result

robust surplus gains depend on **one market property**

with this property

- a rule using only noisy estimates reliably approaches the complete-information optimum
- each $1 spent yields about $2 of market surplus
- consumers essentially unaffected

without it

- in some markets, no rule guarantees even a net gain

the complete-information welfare bound:
$2 of market surplus per $1 spent, with consumers unharmed

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next: attaining this bound from noisy estimates;
then, the mechanism

the authority's information and goal

information

the authority observes noisy demand and quantity estimates

$$\widehat{\boldsymbol{D}} \;=\; \boldsymbol{D} + \boldsymbol{E}, \qquad \widehat{\boldsymbol{q}}^{0} \;=\; \boldsymbol{q}^{0} + \text{noise}$$

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$\boldsymbol{E}$ is symmetric, with independent mean-zero upper-triangular entries on the scale of own-price effects

a **market** is a pair $\theta = (\boldsymbol{D}, \boldsymbol{q}^{0})$, all that welfare depends on

the authority knows a set Θ of possible markets, with no prior

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an *intervention rule* maps the estimates $(\widehat{\boldsymbol{D}}, \widehat{\boldsymbol{q}}^{0})$ to a subsidy vector $\boldsymbol{s}$; it achieves a property ε-**robustly** if

$$\mathbb{P}_{\theta}[\text{property}] \;\geq\; 1 - \varepsilon \quad \text{for every } \theta \in \Theta$$

(such guarantees also hold in expectation)

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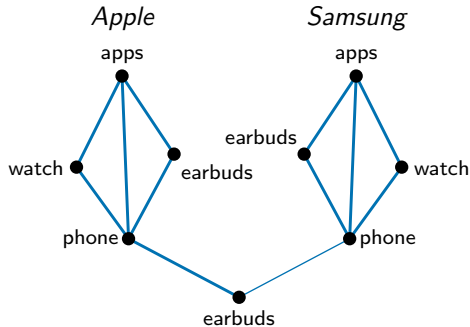
(such guarantees also hold in expectation)

target: the corner of part I, robustly; consumer and producer gains satisfy

$$\Delta\text{CS} + \Delta\text{PS} \;\approx\; 2\,\Delta S, \qquad \Delta\text{CS} \approx 0$$

a dominant principal component of demand

a special case: **two ecosystems**
e.g. Apple and Samsung, each with its own phones, watches, earbuds and apps

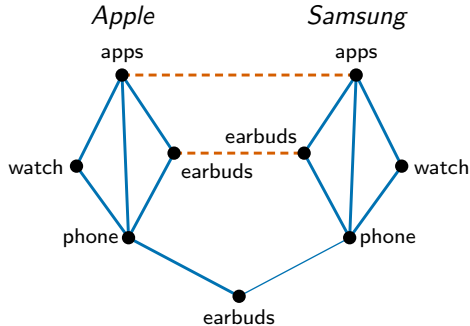


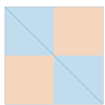
many such goods, $n = 256$

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complements within an ecosystem,
substitutes across





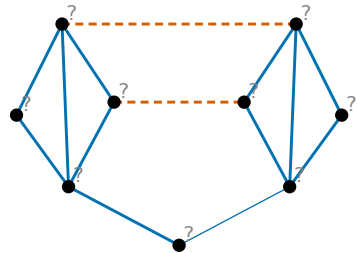
as a demand matrix,
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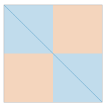
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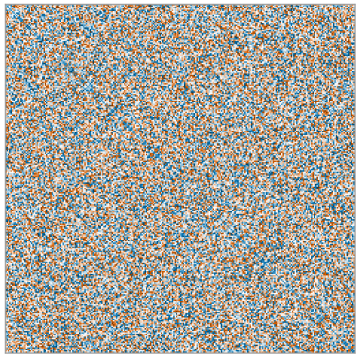
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the data: a noisy 256×256 demand matrix, goods in no particular order

can we recover anything meaningful? what?



observed $\widehat{\boldsymbol{D}}$, arbitrary order

the result for two ecosystems

n goods in two ecosystems;
which good is in which is unknown

within an ecosystem, complements
with cross-price effect c;
across, substitutes

demand estimates with
independent noise on every entry

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Corollary. If $c\sqrt{n} \to \infty$ and the ecosystems' total sales differ, **there exists a rule** that ε-robustly achieves, in large markets:

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complementarity may weaken as the market grows, but more slowly than $1/\sqrt{n}$

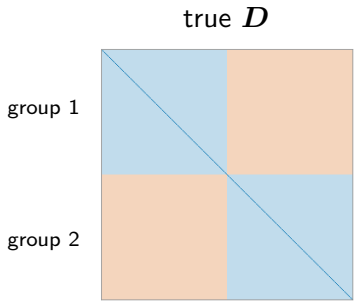
if instead $c\sqrt{n} \to 0$, the method doesn't work (formal statement)

a short film:

recovering structure from noisy data

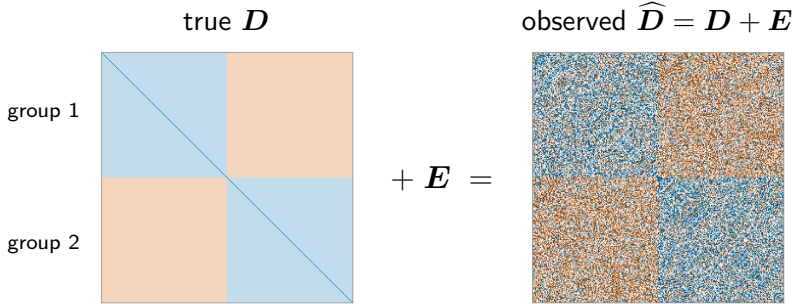
play the film the film as slides after the film →

a planted block structure



negative within groups, positive across; each entry about 0.3 in magnitude

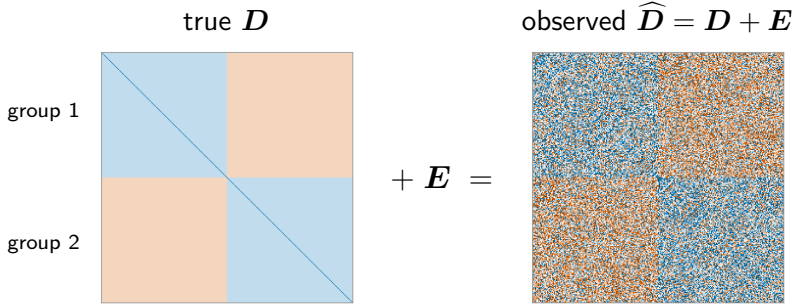
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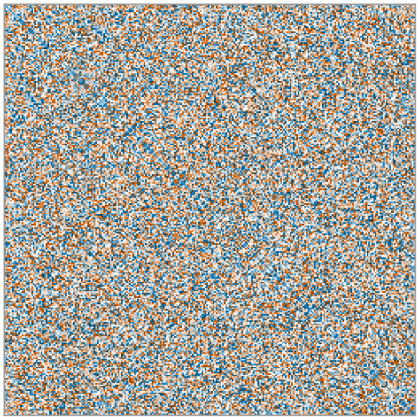


negative within groups, positive across; each entry about 0.3 in magnitude

entry by entry, the noise is about three times the signal

with the goods in the right order, the blocks are faintly visible

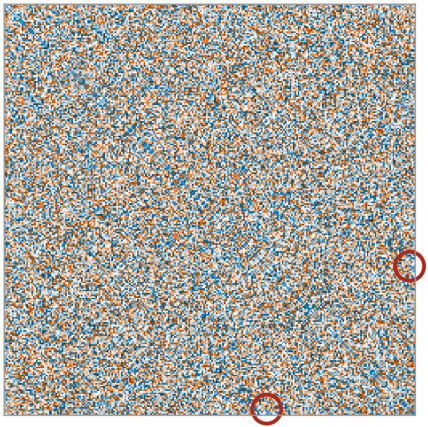
without the labels



observed $\widehat{\boldsymbol{D}}$, arbitrary order

the grouping that makes the picture legible is not known; under an arbitrary ordering of the goods, the pattern is no longer visible

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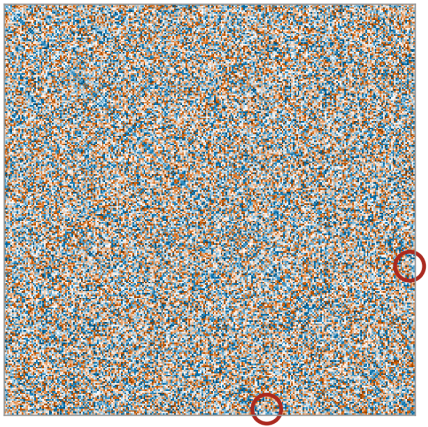


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the sign of an observed entry agrees with the sign of the true entry only **62%** of the time

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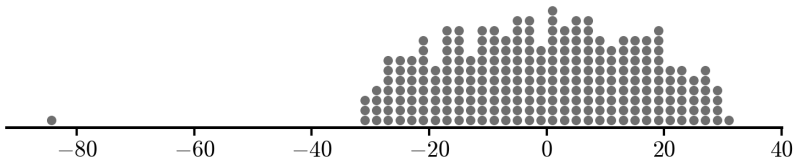
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can we recover the groups, or anything at all?

the spectrum of the observed matrix

compute the eigendecomposition of $\widehat{\boldsymbol{D}}$; in MATLAB, one line:

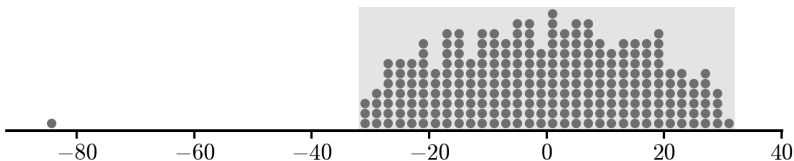
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>> [V, Lam] = eig(Dhat);
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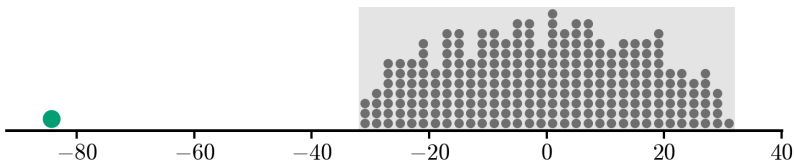


- most of the eigenvalues lie in a **band** around -1, of half-width of order $\sqrt{n}$, set by the noise (the centre is a feature of this example)

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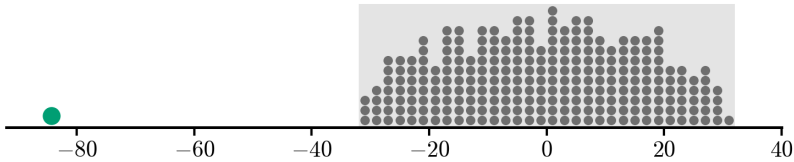


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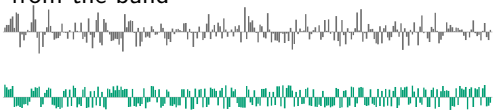
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- one eigenvalue, $\hat{\lambda} = -84$, lies far outside the band
- each eigenvalue has an eigenvector, one entry per good; the outlier's is the so-called **leading eigenvector** $\hat{\boldsymbol{u}}$

the leading eigenvector

from the band

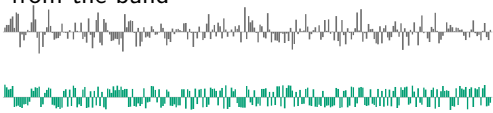


leading, $\hat{\boldsymbol{u}}$

entries in the arbitrary order:
neither shows structure

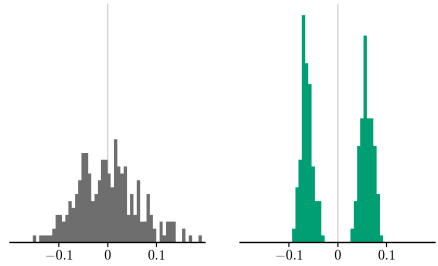
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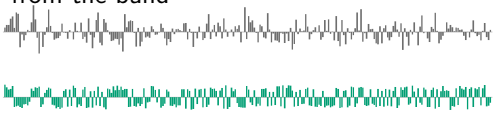


from the band leading

ignore the order and look at the distribution of entries: from the band, concentrated around zero

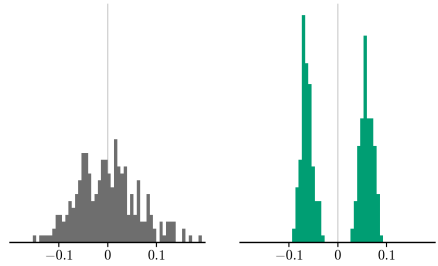
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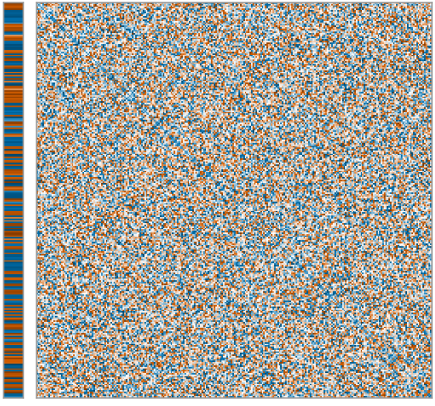
from the band leading

ignore the order and look at the distribution of entries: from the band, concentrated around zero

leading: **two clusters**, one positive and one negative, separated by a gap

two clusters because this example has two groups

assigning goods by sign

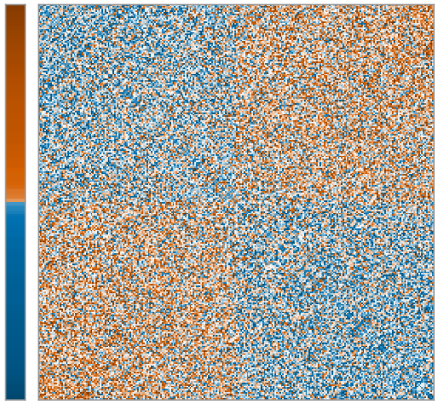


observed $\widehat{\boldsymbol{D}}$, with $\hat{\boldsymbol{u}}$ along the margin

the sign of each entry of $\hat{\boldsymbol{u}}$ can be used to assign its good to a group

the assignment uses only the observed matrix

assigning goods by sign



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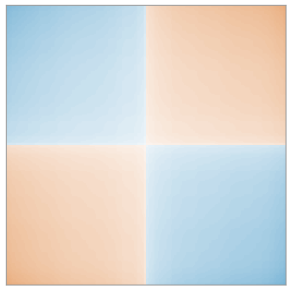
the assignment uses only the observed matrix

ordering the goods by this assignment restores the block structure

all 256 goods are assigned to their true groups

the rank-one term

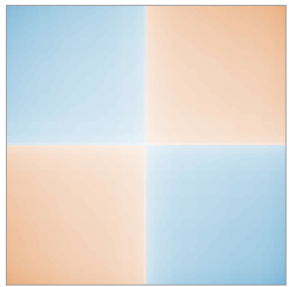
$\hat{\lambda}\,\hat{\boldsymbol{u}}\hat{\boldsymbol{u}}^\top$, from $\widehat{\boldsymbol{D}}$



multiply $\hat{\boldsymbol{u}}$ by itself and scale by $\hat{\lambda} = -84$:
the same two blocks, negative within groups, positive across

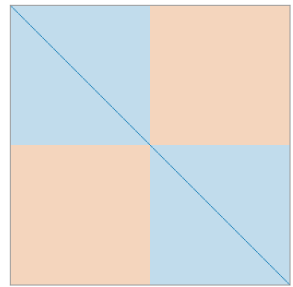
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vs.

true $\boldsymbol{D}$



multiply $\hat{\boldsymbol{u}}$ by itself and scale by $\hat{\lambda} = -84$:
the same two blocks, negative within groups, positive across

the planted structure, estimated from the observed matrix alone

why: the spectral theorem

a symmetric matrix is the sum of rank-one terms, one per eigenvector

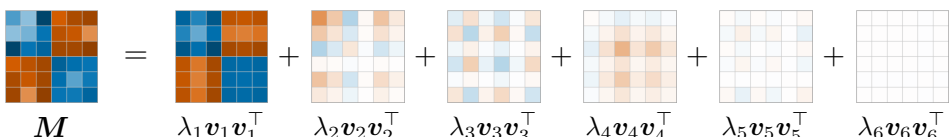
$$\boldsymbol{M} \;=\; \lambda_1 \boldsymbol{v}_1 \boldsymbol{v}_1^\top + \lambda_2 \boldsymbol{v}_2 \boldsymbol{v}_2^\top + \cdots + \lambda_n \boldsymbol{v}_n \boldsymbol{v}_n^\top$$

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six-by-six example: two groups of three, plus noise

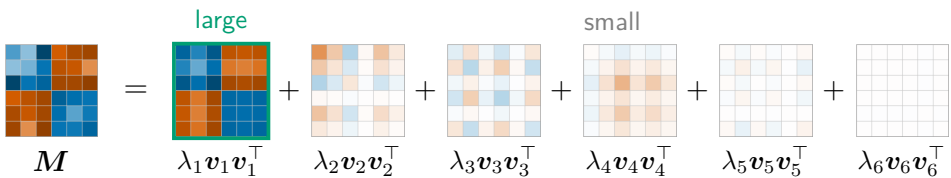


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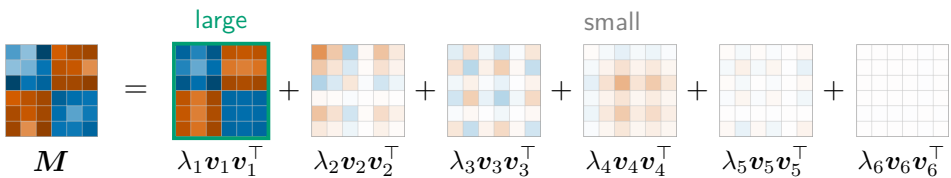
one block makes a much larger contribution, and it alone carries the two groups

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a few much larger terms: random noise cannot change them much

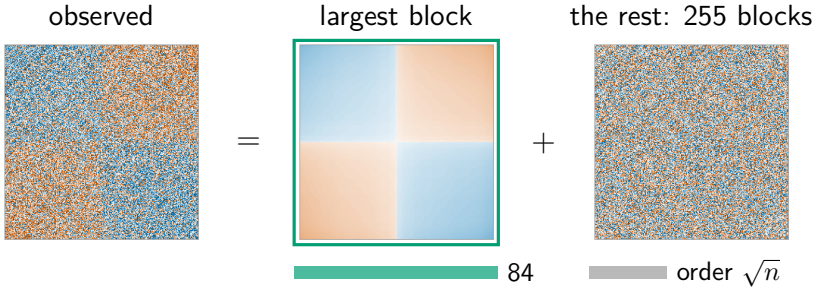
the largest block and the rest

observed | largest block | the rest: 255 blocks



back to the 256 goods: the observed matrix is its largest block plus the rest

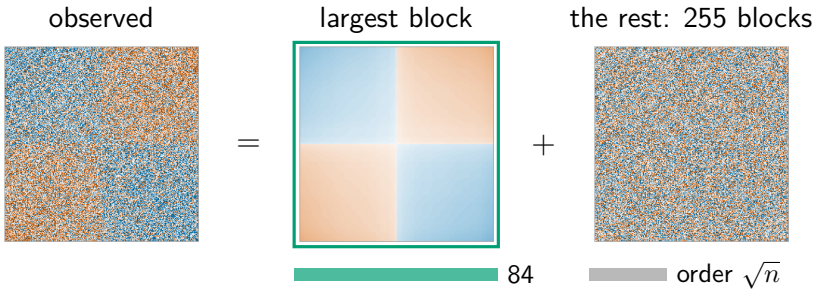
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largest block: eigenvalue -84; the rest, with all the noise: size of order $\sqrt{n}$

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the largest block alone carries the two groups

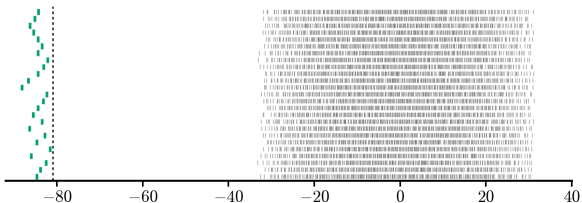
the noise redrawn

the noise changes the largest block in the spectral decomposition of the observed matrix, but only by a bounded amount

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hold $\boldsymbol{D}$ fixed and draw the noise again, 25 times:

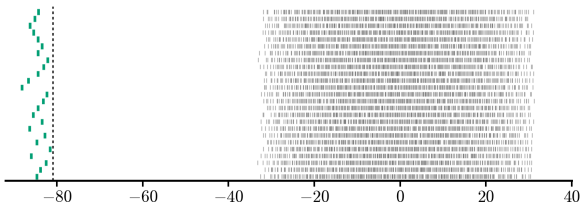


the band rearranges and its eigenvectors change completely;
the outlier moves a few units; the leading eigenvector barely changes

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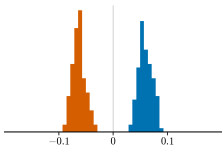
in every draw, no good is misassigned

how strong must the structure be?

reduce the planted eigenvalue, holding the noise fixed

entries of $\hat{u}$; color = true group, not used by the procedure

$|\lambda_1| = 81$



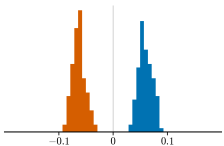
misassigned: 0

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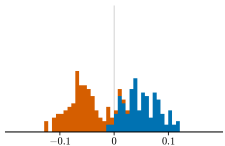
entries of $\hat{u}$; color = true group, not used by the procedure

$|\lambda_1| = 81$



misassigned: 0

$|\lambda_1| = 33$



misassigned: 9

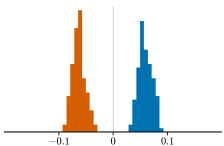
at 33, nine goods are misassigned

how strong must the structure be?

reduce the planted eigenvalue, holding the noise fixed

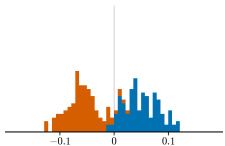
entries of $\hat{u}$; color = true group, not used by the procedure

$|\lambda_1| = 81$



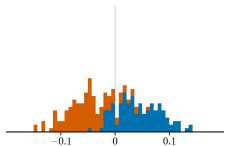
misassigned: 0

$|\lambda_1| = 33$



misassigned: 9

$|\lambda_1| = 25$



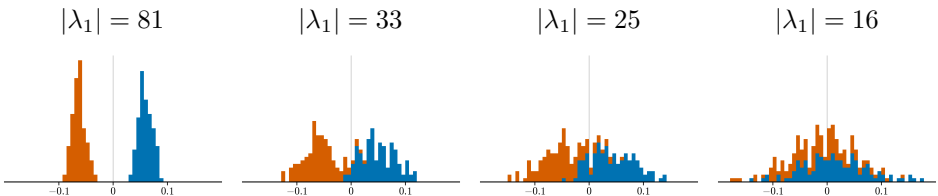
misassigned: 35

at 25, the clusters begin to merge

how strong must the structure be?

reduce the planted eigenvalue, holding the noise fixed

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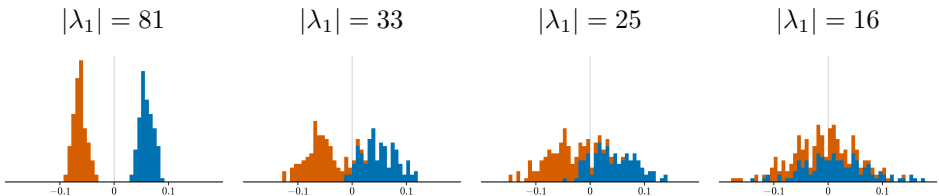


at 16, the eigenvalue lies inside the band: its eigenvector carries no information about the groups

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misassigned: 0 | misassigned: 9 | misassigned: 35 | misassigned: 104

at 16, the eigenvalue lies inside the band: its eigenvector carries no information about the groups

structure weaker than the noise is not recovered; structure sufficiently stronger is

back to demand

nothing here uses the economics; it is a property of symmetric matrices

a planted component that is large relative to the rest is recovered by its eigenvector

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back to demand

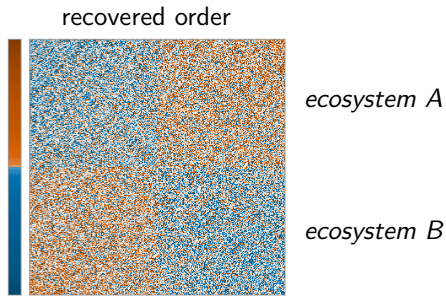
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in the model, the planted component is the **ecosystem structure** of demand

the leading eigenvector is what the intervention rule uses

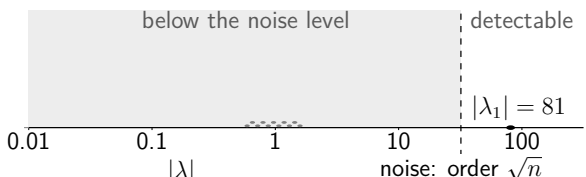


significant structure

significant structure requires detectability and quantity relevance:

1. some eigenvalues are far larger than the noise level, of order $\sqrt{n}$

$$|\lambda| \geq b(n), \quad \|\boldsymbol{E}\|/b(n) \to 0$$



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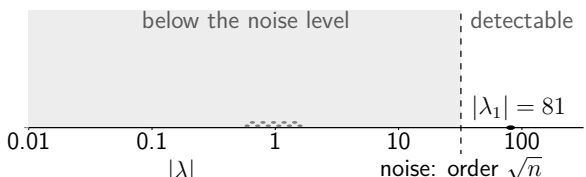
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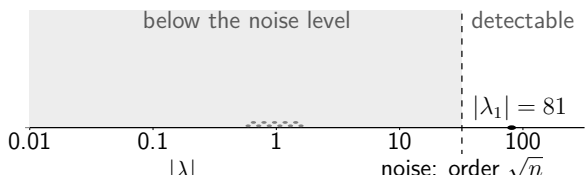
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economic interpretation: large-scale complementarity

also: quantity noise no larger than quantities

main result

Theorem 1. Under significant structure, for any $\varepsilon > 0$ and budget $\bar{S}$, **there exists a rule** that ε-robustly achieves, in large markets:

the math →

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1. market surplus of about \$2 per dollar

$$\Delta\mathrm{CS} + \Delta\mathrm{PS} \;\geq\; (2 - \varepsilon)\,\Delta S$$

the math →

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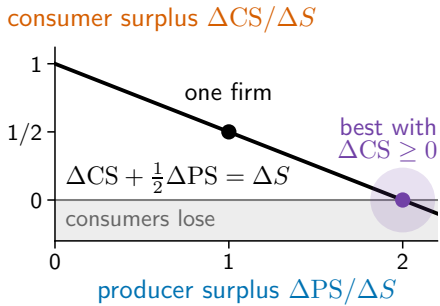
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the complete-information **corner**

first-order effects; regularity: backup

the math →

the rule

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the math →

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2. project observed quantities $\widehat{\boldsymbol{q}}^0$ onto $\widehat{\mathcal{L}}$
3. scale the subsidy vector to budget $\bar{S}$

$$\boldsymbol{s} \;=\; \bar{S}\,\frac{\mathbb{P}_{\widehat{\mathcal{L}}}\,\widehat{\boldsymbol{q}}^0}{\|\mathbb{P}_{\widehat{\mathcal{L}}}\,\widehat{\boldsymbol{q}}^0\|^2}$$

no inversion of $\widehat{\boldsymbol{D}}$

projection averages over many noisy entries

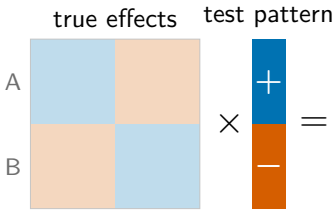
next: intuition and proof

the math →

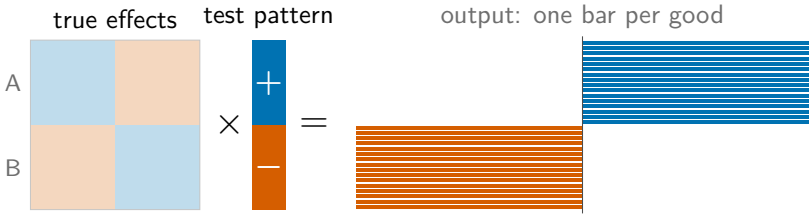
part III:

economic intuition

structure and noise



structure and noise

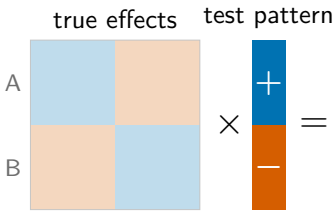


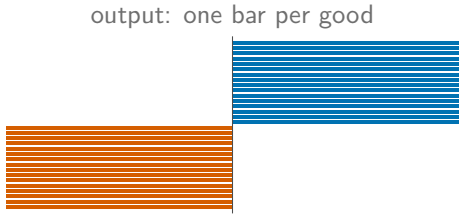
adds up

each term $+0.32$

256 terms: about 81

structure and noise

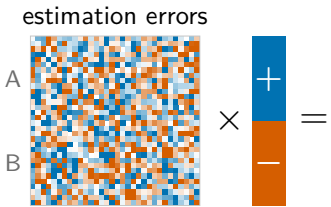




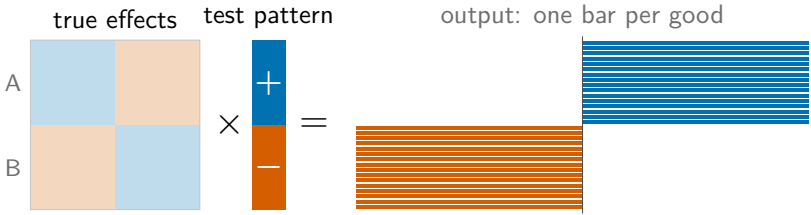
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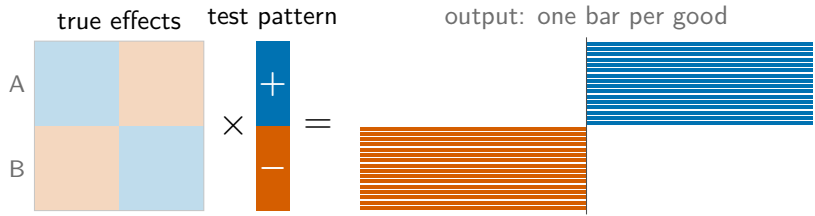
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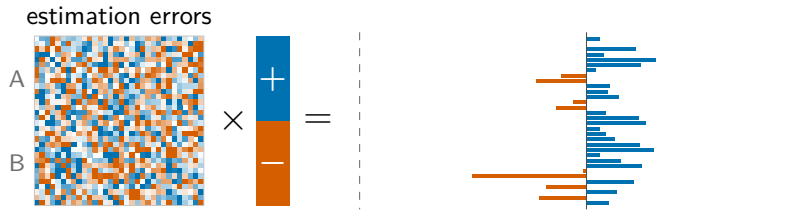
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with n goods, structure grows like n, noise only like $\sqrt{n}$

spectral price theory

demand, written in its eigenvectors (the spectral theorem, as in the film):

$$\boldsymbol{D} \;=\; \lambda_1 \boldsymbol{u}^1 (\boldsymbol{u}^1)^\mathsf{T} + \lambda_2 \boldsymbol{u}^2 (\boldsymbol{u}^2)^\mathsf{T} + \cdots + \lambda_n \boldsymbol{u}^n (\boldsymbol{u}^n)^\mathsf{T}$$

each eigenvector $\boldsymbol{u}^\ell$ is a **bundle** of goods; $|\lambda_\ell|$: how strongly its sales respond to its price

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barely responds

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quantity $\boldsymbol{u}^\ell \cdot \Delta \boldsymbol{q} = \dfrac{|\lambda_\ell|}{1+|\lambda_\ell|}\, \boldsymbol{u}^\ell \cdot \boldsymbol{s}$

welfare $\Delta W = \displaystyle\sum_\ell \frac{|\lambda_\ell|}{1+|\lambda_\ell|}\, \Delta S_\ell$

$\Delta S_\ell = (\boldsymbol{u}^\ell \cdot \boldsymbol{q}^0)(\boldsymbol{u}^\ell \cdot \boldsymbol{s})$, spending on bundle ℓ

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the math →

subsidy incidence

per $1 of subsidy	**isolated good** $\lvert\lambda\rvert = 1$
price reduction	 50¢
consumer surplus	 50¢
producer surplus	 $1.00
market surplus consumers + producers	 $1.50

subsidy incidence

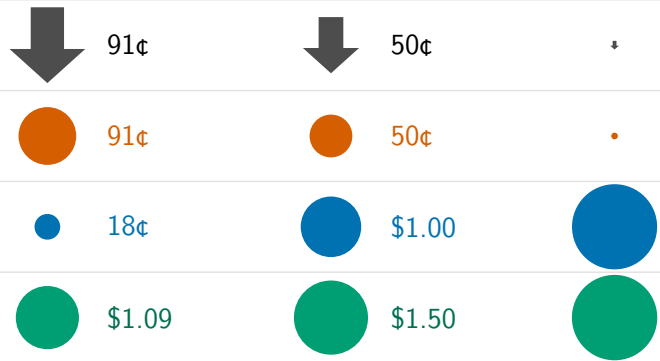
per $1 of subsidy	**weak demand response** $\lvert\lambda\rvert = 0.1$	**isolated good** $\lvert\lambda\rvert = 1$
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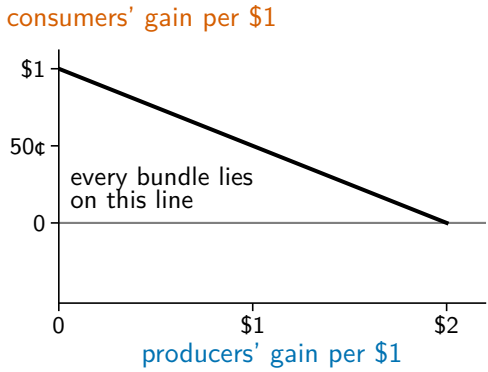


little pass-through, output expands: about **$2 of market surplus** per $1

surplus, eigenvector by eigenvector

any subsidy is a mix of bundles; its gain per \$1 averages the bundles' gains, weighted by spending

each bundle lands on the line of Proposition 1; its eigenvalue says where



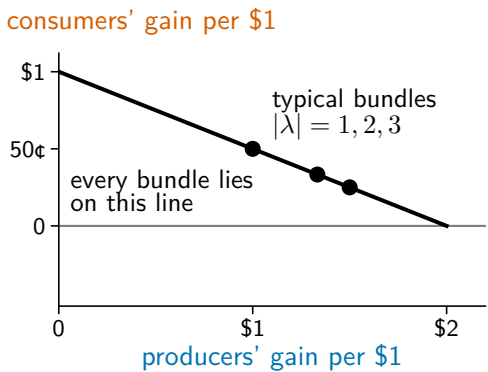
the math →

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the math →

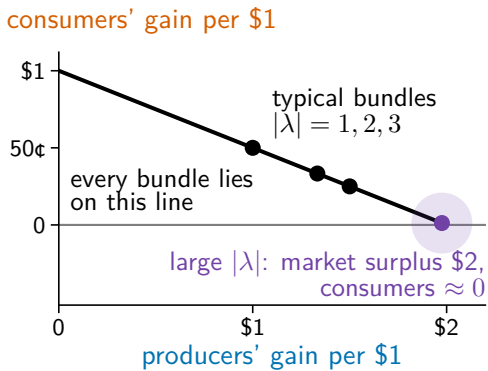
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about \$2; consumers about 0: the corner



the math →

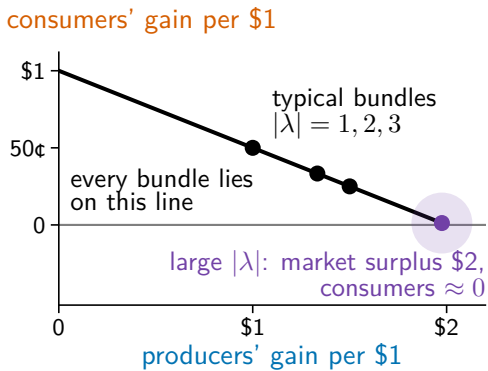
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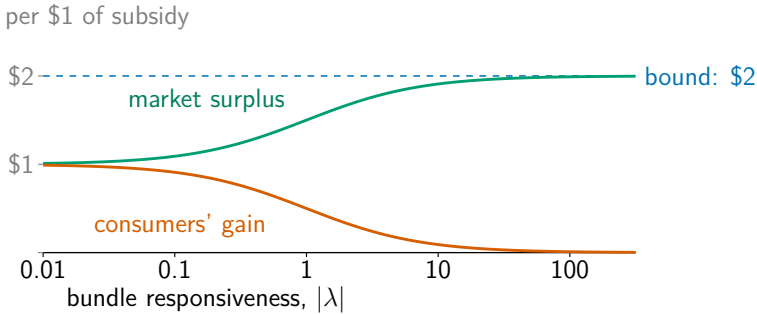
bundles with **large** $|\lambda|$:
about \$2; consumers about 0: the corner



to reach the corner, spend only on bundles with large $|\lambda|$, if we can find them

the math →

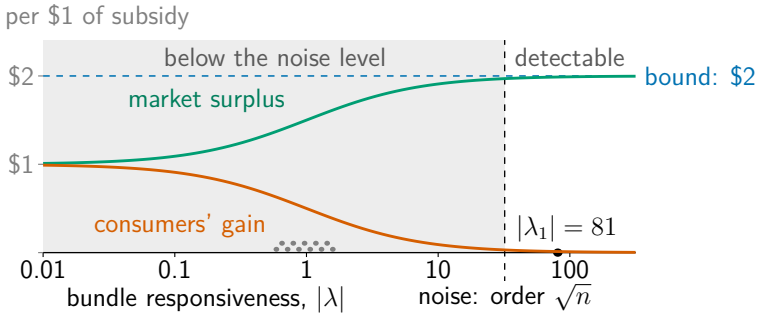
recovery and surplus



surplus: per $1, strong patterns yield about $2 of market surplus, consumers about 0

the math →

recovery and surplus

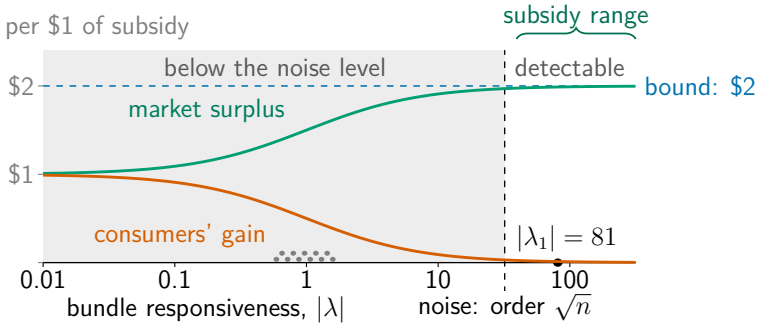


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recovery: only strong patterns stand out from the noise

the math →

recovery and surplus



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recoverable directions yield the largest subsidy gains

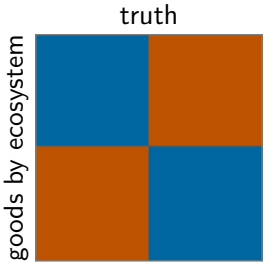
the math →

part III, continued:

formal argument

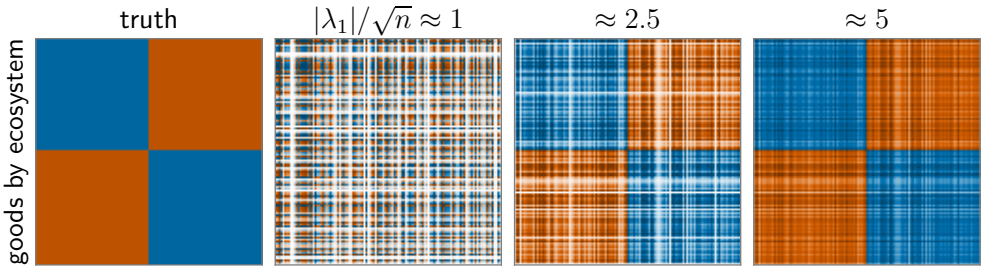
Weyl and Davis–Kahan (reserve) $\rightarrow$

recovering structure



blue: complements vermillion: substitutes

recovering structure



estimated matrix $-\widehat{\boldsymbol{u}}^1(\widehat{\boldsymbol{u}}^1)^\top$ as $|\lambda_1|$ increases:

noise $\rightarrow$ partial recovery $\rightarrow$ distinct groups

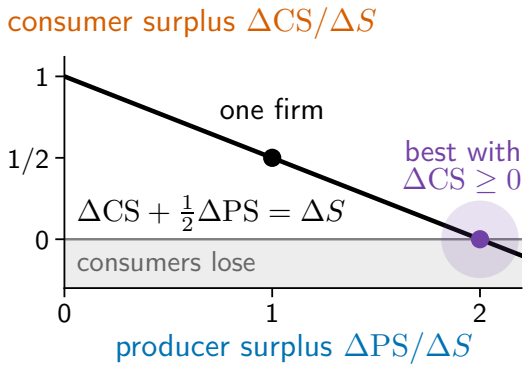
complementary groups are recovered without estimating every entry

surplus by eigenvector

Lemma. A \$1 subsidy along a demand eigenvector with eigenvalue λ affects only that direction:

consumer surplus $\dfrac{1}{1+|\lambda|}$

market surplus $\dfrac{1+2|\lambda|}{1+|\lambda|}$



the math →

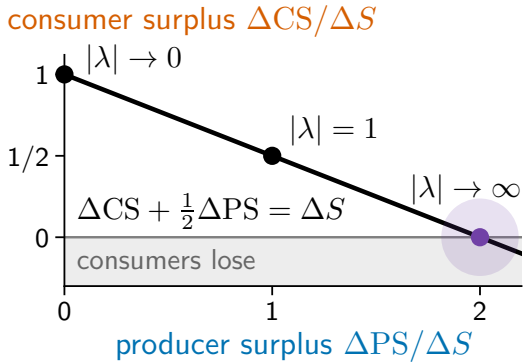
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a 1¢ price cut increases sales by $|\lambda|$ units; large $|\lambda|$ implies low pass-through and greater output



the math →

surplus by eigenvector

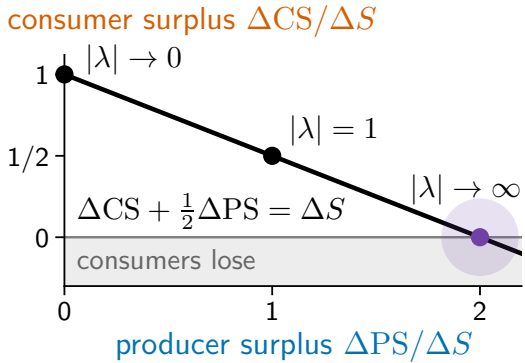
Lemma. A \$1 subsidy along a demand eigenvector with eigenvalue λ affects only that direction:

consumer surplus $\dfrac{1}{1+|\lambda|}$

market surplus $\dfrac{1+2|\lambda|}{1+|\lambda|}$

a 1¢ price cut increases sales by $|\lambda|$ units; large $|\lambda|$ implies low pass-through and greater output

large $|\lambda|$ suffices for near-maximal surplus



the math →

proof outline

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Theorem 1 follows

the rule in the experiment

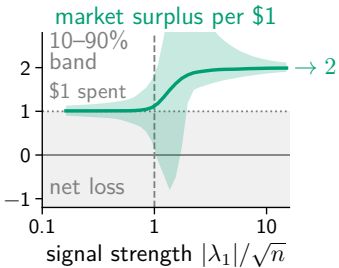
$$\boldsymbol{D} \;=\; -|\lambda_1|\,\boldsymbol{u}^1(\boldsymbol{u}^1)^\top \;+\; \underbrace{\text{aggregate}}_{|\lambda|=0.01} \;+\; \underbrace{\text{bulk}}_{|\lambda|\approx 1}$$

256 goods, two ecosystems, 500 noise draws per $|\lambda_1|$

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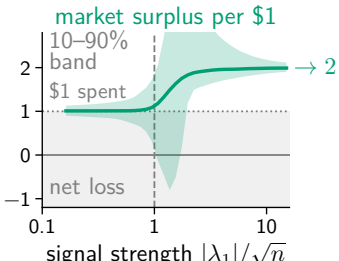
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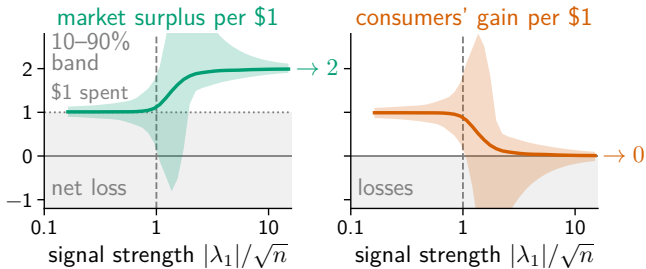
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market surplus → $2

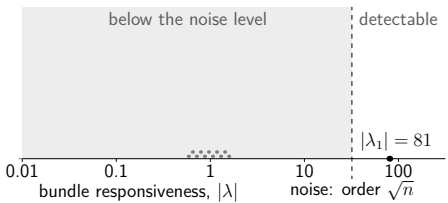
consumers' gain → 0

the limits of robust interventions

Proposition 3, for rules with $\Delta S > 0$:
1. in some markets **without** significant structure, no rule ε-robustly achieves

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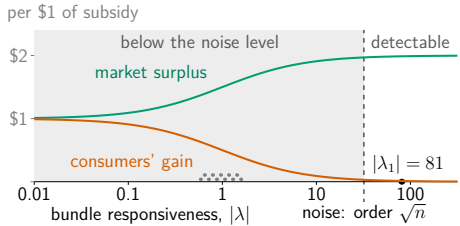
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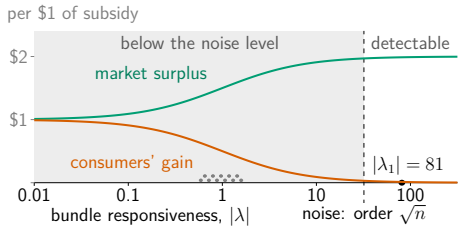
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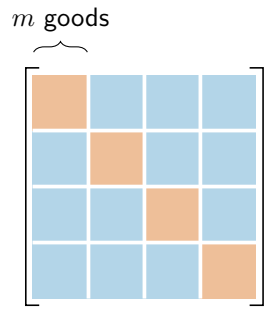


these limits follow from noise at small eigenvalues

the math →

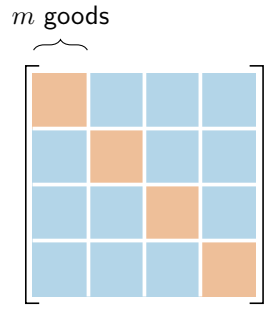
sources of significant structure

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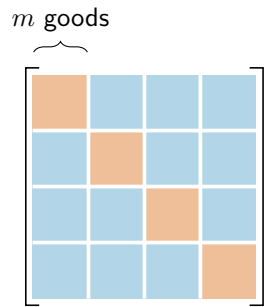
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- all-substitute (hedonic) demand models lack significant structure Pellegrino (2021)



substitutes within

complements across

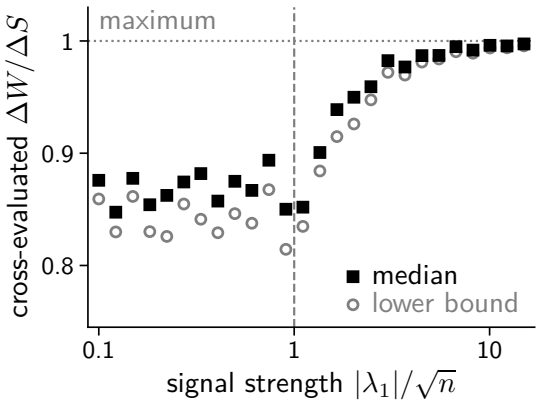
K categories of m goods

a diagnostic for significant structure

- estimate the rule on one data sample
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- evaluate its net gain per dollar, $\Delta W/\Delta S$, on an independent sample
- significant structure $\Rightarrow$ ratio ≈ 1
- one-sided: success certifies; failure is inconclusive



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two paths:

- direct oligopoly theory: recover the curvature too
- black-box pass-through: **compressed sensing**

other extensions: network games

our pricing game is a network game with linear best replies:
the network is the complements and substitutes among goods

other games with noisily known spillovers:

- public goods: align private and social returns
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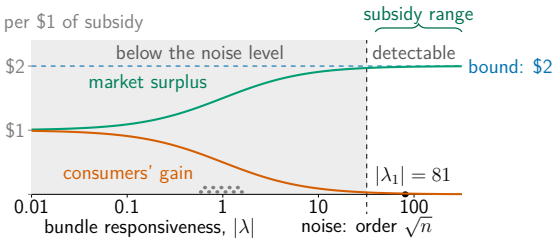
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we used consumer demand theory in a non-trivial way:
the Slutsky matrix is symmetric and negative semidefinite
not a straightforward extension, but a very interesting one

recap



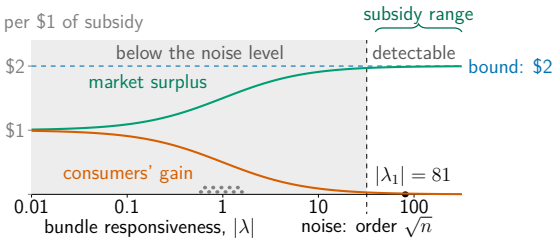
bound: the corner, \$2 of market surplus per \$1, consumers unharmed (Prop. 1)

recovery: $|\lambda| \lesssim \|\boldsymbol{E}\|$ is unresolved; above it, recover the top eigenspace (Weyl, Davis–Kahan)

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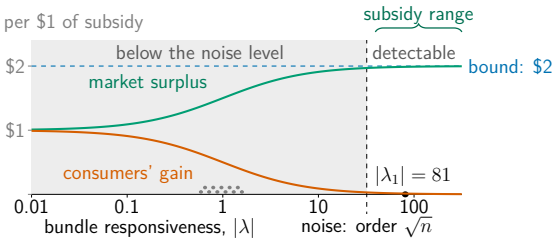
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recoverable eigendirections yield the largest subsidy gains

conclusion

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consistent estimation
in small markets

large markets

economically important, and large enough
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other applications, e.g. the macroeconomics of networks:
noisy measurements of individual links, but structure that transcends them